Magnetic Reconnection and the Blandford-Znajek Process

Energy Extraction Mechanisms For AGN

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When it comes to the study of AGN and the launching of relativistic jets, it is a field of study which has evolved much over the last few decades. That evolution has given rise to the most commonly cited explanation for the energy extraction mechanism which powers these jets: the Blandford-Znajek (BZ) process, a version of the more general Penrose Process. The functionality of the BZ mechanism relies on the twisting of magnetic field lines to create a collimated Poynting Flux to carry the rotational energy of a supermassive Kerr Black Hole outwards from its Event Horizon and into the space surrounding it, generating jets. Whereas previous models using this energy extraction mechanism assumed the outward energy transport came from the coiling of magnetic fields and the eventual release of that energy as if a spring was expanding, recently theories on the role of magnetic reconnection (MR) in the process have surfaced in the AGN community. In this review, we look at the general Blandford-Znajek process and compare it to an alternative energy extraction framework centered around MR, analyzing their respective efficiencies, contrasting extraction processes, and how what implications each mechanism has on the existence of the other.

1 Introduction

Humanity's desire to explore beyond our terrestrial existence has long been a defining aspiration. As we continue to make great leaps in astronomy and astrophysics, we proceed in our journey of asking the existential questions of where everything came from and where it is all heading. Among the many known celestial bodies, Active Galactic Nuclei (AGN), such as quasars and blazars powered by supermassive black holes (SMBHs) (Böttcher 2019; Lalakos et al. 2025), are especially valuable for understanding the phenomena which shaped the early universe and continue to evolve it today. From their accretion disks (AD) to their relativistic jets (RJ), this area of research can give us much insight into how the universe operates (Blandford and Znajek 1977; Kinoshita and Igata 2018; Comisso and Asenjo 2021). For example, understanding the process of AGN feedback has been key to explaining the regulation of star formation efficiency in massive galaxies (Harrison and Ramos Almeida 2024).

One evolving area of AGN research surrounds the launching of the aforementioned RJs, which propagate at relativistic velocities (Böttcher 2019; Lalakos et al. 2025). While there have been many attempts to explain why these jets move at such speeds and why they form in the first place, there currently isn't a completely precise answer on what process, or

combination of processes, end up producing these extremely bright phenomena (Böttcher 2019). Specifically, the main question about these jets is where they get all their energy from in order to be launched at such high speeds in the first place (Blandford and Znajek 1977; Böttcher 2019). It can be found that a commonality between several prominent theories is the key role of magnetic fields in the energy extraction and transport mechanisms of black holes (BH) systems (Koide and Arai 2008). Assuming that all AGN are powered by Kerr BHs (ones with angular momentum), whether they have jets or not, several of the posited extraction mechanisms using a magnetic field explanation become interesting due to their unique reliance on angular momentum transport to create an outward energy flux, also known as the Poynting Flux (PF) (Blandford and Znajek 1977; Zweibel and Yamada 2016). Attempts to blend these explanations, such as the Blandford-Znajek (BZ) mechanism, with observations of simulations of AGN, have resulted in modern interpretations that suggest there exists certain conditions which lead to the efficient launching of RJs, such as a magnetically saturated AD strongly regulating the inflow of accreted matter, adding a nuance to the triggering of jets in AGN (Blandford and Znajek 1977; Tchekhovskoy et al. 2011). Despite this, the question of why some AGN have jets and why others don't remains a mystery that has yet to be solved, representing one of the loose ends of multiple theories on the mechanisms driving RJs, as mentioned earlier (Fabian 1999). In the end, despite the questions that still surround this branch of astrophysics, much progress has and is still being made on how and why AGN have jets and in possible correlations between RJ formation and accretion patterns (Shen and YuChih 2025; Beedle et al. 2025; Böttcher 2019). In this review, we examine the foundation for the energy extraction mechanisms covered by this review paper. From there, we will extrapolate the necessary principles to be applied in explaining one of the most prominent explanations for RJs in AGN: the BZ process. Finally, we will use the same groundwork to briefly delve into and discuss the implications of magnetic reconnection (MR) on jet formation, later focusing on comparitive analysis between the two mechanisms as well as potential future prospects in this field of research.

2 Methods

This literature review was done using Google Scholar, arXiv, and ResearchGate to find papers on the BZ process, MR, and all relevant information relevant to the understanding of those topics. This means that most papers were evaluated and used for the literature review only if their contents were directly discussing the BZ mechanism and how it works. For the papers that had no direct mention of the BZ process, these were only selected in order to provide a high-level understanding of a concept which was key to the BZ mechanism, but here too it was avoided to go too deep into the research around fringe topics.

In terms of the keywords and phrases used when finding papers, most of the time it came down to using basic key words like "AGN jets", "Blandford-Znajek Mechanism" (or "Process"), "Black hole energy extraction Kerr", "Magnetic reconnecting in AGN", "Force-Free Magnetosphere AGN" and "AGN Energy Extraction Negative Energy At Infinity".

3 Discussion

3.1 Black Hole Accretion Framework

Black Holes are defined by their immense mass at the singularity and the Event Horizon (EH) which surrounds it. Due to this immense mass, a hallmark of a BH, whether rotating or not, is its incredibly powerful gravitational strength, especially at and past its EH, where not even light can escape its pull. For BHs on the Kerr metric, or BHs which rotate and have angular momentum, the rotation of the celestial body twists the space-time around it as it continues to spin in a process known as "frame-dragging" (Ricarte et al. 2022). In fact, the "stationary limit" of a rotating BH is the boundary past which nothing, not even light, is able to stay still and must rotate with the BH: this region of extremely powerful frame-dragging is called the ergosphere, and this is where the rotational energy of a BH is extracted in order to power RJs (Blandford and Znajek 1977; Kinoshita and Igata 2018; Ricarte et al. 2022). As an object moves deeper into the ergosphere and closer to the EH, the ergosphere's frame-dragging effect becomes more potent, a concept that will become more important later on in this paper (Ricarte et al. 2022).

In addition to their ability to frame-drag, BHs on the Kerr metric (and really in general) are characterized by their ability to pull in matter and grow in size at some time or another. This process is called accretion, and the rate at which a BH accretes material is given by the Bondi Accretion Rate

$$\dot{□}_{□} = □□□_{□}^{2}\, □_{0}\, □_{□},$$

where ρ_0, $□_{□}$, and $□_{□}$ are defined as density, the speed of sound, and the Bondi radius, respectively (Lalakos et al. 2025). $□_{□}$ is further defined by Eq. [radiusEq].

$$□_{□} = \frac{□□_{□□}}{□_{□}^{2}},$$

Here $□$ is the gravitational constant and $□_{□□}$ is the BH mass (Lalakos et al. 2025). While the object feels a gravitational pull towards the singularity of the BH, it also inherently carries momentum of its own, allowing it to "orbit" the BH (Shakura and Sunyaev 1973; Blandford and Znajek 1977). This condition holds true as long as the object moves at speeds fast enough to avoid it falling into the BH and being accreted; with particles on a mass scale, this forms what is called an accretion disk (AD) (Shakura and Sunyaev 1973; Blandford and Znajek 1977). Out of the particles constituting the accretion disk, many are magnetized, and support a stationary magnetosphere (which will be explained later) when moving throughout the AD (which creates a current) (Blandford and Znajek 1977; Komissarov 2004).

At the same time, it is key to note that ADs follow Keplerian rotation (Blandford and Znajek 1977), which is to say that the disk's rotational speed decreases as it moves farther from a central, dominant mass (or the BH) (Shakura and Sunyaev 1973). What this tells us is that particles deeper in the AD rotate at a faster speed than those farther out, creating differential rotation of particles (Shakura and Sunyaev 1973; Oishi et al. 2020). However,

considering Ferraro's law of isorotation, we know that the angular velocity of a plasma (such as AD) is constant across a field line, such as those of the magnetic field (Ferraro 1937). As a result of the two principles outlined above, the differential rotation causes the magnetic field lines to stretch and twist in the AD (Lovelace and Romanova 2003; Balbus and Hawley 1991). The direct consequence of this is that the magnetic tension of the field line ends up straightening the field line (since that's what tension does) to keep the magnetic field in a lower energy state (Myers et al. 2017; Balbus and Hawley 1991). Given that magnetic tension is proportional to both magnetic field strength and magnetic flux through an area, these variables are correlated and this relationship will be cited several times throughout the rest of the paper.

When weaker magnetic field lines thread the AD and are "corrected" and "straightened" by magnetic tension, what ends up happening is that they "over-correct" the kinks in the field, destabilizing the AD by forcing regions of the disk to rotate too quickly for their position (Oishi et al. 2020; Balbus and Hawley 1991). Angular momentum and energy is transported towards the ends of the AD, a process known as a Magnetorotational Instability (MRI), or a Balbus–Hawley instability (Oishi et al. 2020; Balbus and Hawley 1991). The fundamental working of an MRI is that two regions within the AD are coupled to magnetic field lines (Oishi et al. 2020; Balbus and Hawley 1991). With the previous assumptions made about Ferraro's Law, any plasma along the aforementioned field line is forced to rotate with it, and so when that field line is overcorrected, this launches plasma along the line with much more angular momentum than expected, destabilizing the AD (Oishi et al. 2020; Balbus and Hawley 1991). As a result of this, corotation of the plasma along the magnetic field line is not achieved, but instead turbulence is induced in the AD which leads to a turbulent dynamo effect which strengthens the magnetosphere of the AD (Komissarov 2004; Kunz et al. 2016).

The strengthening of the magnetic field also creates a stronger electric field surrounding the BH, to the point where a force-free condition is developed, a key underlying quantum electrodynamics assumption made when describing the energy extraction mechanisms for AGN (Komissarov 2004, 2002). This state is facilitated by the Schwinger effect (Blandford and Znajek 1977; Iarygina et al. 2026), where extremely powerful electric field spontaneously creates electrons and positrons (in a process known as pair production), which race along field lines to cancel out the electric field. This leaves the magnetic field as the only structure which governs the motion of charged particles (Blandford and Znajek 1977; Komissarov 2004).

Having established this background, the rest of the BZ process can now be explored in how it functions. As mentioned before, MRIs rely on weak magnetic fields to exist in order to operate. This is because the "correction" from the magnetic tension moves at the speed of the Alfvén wave (v_A), which is directly proportional to the strength of the magnetic field (B), as given by the equation

$$v_A = \sqrt{\frac{B^2}{(\mu_0 \rho)}}$$

where μ_0 is the permeability of free space and ρ represents the plasma density (Sabri et al. 2020). Given that the above statements are true, the fundamental reason that MRIs are able to develop in AD is because the Alfvén wave doesn't move through the magnetic field at a fast enough speed to correct any kinks in the field before an MRI develops (Blaes et al. 2025; Balbus and Hawley 1991). However, this relation also tells us that as the magnetic field strength increases, the Alfvén wave should speed up and eventually be able to straighten out field lines before the twists end up inducing an instability (Blaes et al. 2025). Hence, as the BH accretes more and more matter and charged particles, its magnetic field strength slowly increases over time until it reaches a state where its accretion disk is saturated with magnetic field flux, a condition known as a "Magnetically Arrested Disk" (MAD) (Liska et al. 2020; Lovelace and Romanova 2003).

In the MAD state, the magnetic field strength of the AGN is at its max, meaning that the tension is at its strongest and the Alfvén wave is at its fastest, therefore meaning that magnetic tension is able to shut down MRIs before they develop in the first place (Liska et al. 2020; Balbus and Hawley 1991; Blaes et al. 2025). What this looks like intuitively is essentially the magnetic tension no longer "overcorrecting" bends in the field, and instead simply transporting angular momentum outwards to maintain stable corotation of the magnetic field across the AD in order to maintain straight magnetic field lines and a low-energy geometry (Ferraro 1937; Balbus and Hawley 1991; Blaes et al. 2025). This idea of a more rigid geometry is ultimately what leads to the energy extraction mechanism behind the BZ process being most efficient, and as such, the MAD state has been observed to be heavily correlated to such energy extraction mechanisms powering RJs (Liska et al. 2020; Lalakos et al. 2025).

3.2 The Blandford Znajek Process

Going off of the previously established idea that the MAD state enforces corotation of the magnetic field around the BH through Alfvén waves and magnetic tension, we simply need to understand one last fundamental concept before we can finally look at how the Blandford-Znajek process actually works (Kinoshita and Igata 2018). It is key to note that the magnetic field geometry is based off of the orientation and angular velocity of the plasma in the AD. This means that if the plasma is rotating with angular velocity $\Omega_{\square}$, then its magnetic field's geometry will naturally (with no other factors considered) be still within that rotating frame of reference, which forms the foundation of the stationary magnetosphere (Komissarov 2004).

Understanding this, as well as how Alfvén waves as well as magnetic tension interact with MRIs and MAD states, a clear picture of the functionality for the BZ process can be illustrated. As described above, the effects of the ergosphere become more potent the closer you are to the source of the frame-dragging, which is the SMBH at the center of the AGN, in this case. Seeing as the corotation of the magnetic field follows a corotation vector, as the magnetic field lines get closer and closer to the SMBH singularity, they are being frame-dragged more and more (Blandford and Znajek 1977; Kinoshita and Igata 2018). The result of this is that more "inner" parts of these field lines have to move faster in order to keep up with the corotation vector of the magnetic field (Kinoshita and Igata 2018), since the geometry and corotation of the magnetic field has already been established by the MAD

state and the nature of the AD (as described previously). Using this line of reasoning, there exists a "Light Surface" (LS) at which the frame-dragged object must move at a speed equivalent to that of the speed of light in order to keep up with the corotation vector (Blandford and Znajek 1977; Kinoshita and Igata 2018). While this is possible for the magnetic field, since it has no mass, the problem occurs past the LS, where the object would have to move at speeds faster than that of light to keep up with the corotation vector (Kinoshita and Igata 2018; Komissarov 2004). Essentially, beyond the LS, the corotation vector becomes spacelike and no longer can be kept up with while obeying time (Kinoshita and Igata 2018; Komissarov 2004).

Cross-applying this idea to the magnetic field rotating, it means that the inner parts of the magnetic field, specifically the components past the LS, cannot corotate with the rest of the magnetic field, stretching and twisting the magnetic field lines (Kinoshita and Igata 2018; Komissarov 2005). This contortion is compounded by the fact that the parts of the magnetic field outside of the LS are rigidly rotating in geometries because of the field configurations described previously. This bending and stretching of the magnetic field lines ends up storing energy within the field lines in the form of the tension of the field line as the field lines continue to twist more (Kinoshita and Igata 2018; Komissarov 2005).

The twists, as disturbances, are then propagated outwards through electromagnetic stresses in the form of Alfvén waves (Kinoshita and Igata 2018; Sabri et al. 2020). It is key to note that the angular momentum and energy are transported by electromagnetic stresses (driven by magnetic tension), and thus are also propagated outward by the Alfvén wave (Kinoshita and Igata 2018; Komissarov 2005; Lalakos et al. 2025). As such, angular momentum is being redistributed outwards through the magnetic field as the BH's rotational energy is transported out of the SMBH over time through the support of magnetic tension: something known as the PF (Blandford and Znajek 1977; Kinoshita and Igata 2018; Komissarov 2005). Specifically, this energy extraction is performed along the direction of the magnetic field line pointing away from the EH of the SMBH (Kinoshita and Igata 2018). This forms the basis for the energy extraction present in the BZ mechanism by effectively extracting the SMBH's angular momentum and energy, to power the RJs, through the ergosphere's frame-dragging of the magnetic field.

While keeping the above in mind, one of the last effects of frame-dragging must be considered: the toroidal magnetic field. By its nature, magnetic field lines run from one pole to another. However, as the field lines stretch further into the ergosphere and closer to the EH, the magnetic field lines start to become more and more frame-dragged, causing the field lines to take on a more torus-like shape, forming a toroidal component of the magnetic field (Kinoshita and Igata 2018; McKinney 2006). This toroidal region is what is being twisted by the mismatched corotation of the magnetic field, thus extracting energy from the BH (Kinoshita and Igata 2018; McKinney 2006). On the other hand, the poloidal magnetic field dictates the direction in which the PF goes. Due to the conic shape that the poloidal field lines end up making, they form a sort of "funnel", through which the flux is concentrated into a small space (Kinoshita and Igata 2018; McKinney 2006). As a result, the particles present in that small space are collimated and accelerated to high speeds, creating the phenomena of the RJs (Blandford and Znajek 1977; Kinoshita and Igata 2018; McKinney 2006).

3.3 Negative Energy At Infinity

To understand what MR involves, the BZ mechanism must first be looked at through a different lens. More specifically, the Blandford-Znajek process must be compared more directly to the Penrose Process, in that it extracts rotational energy from a SMBH through "negative energy" falling into the EH and positive energy escaping (Piran and Shaham 1978; Koide and Arai 2008). To explain this, we will take the ergosphere and consider its effects on phenomena. Within this region, angular momentum and energy are tightly coupled (Grib and Pavlov 2013), and to an observer at an infinite distance away (henceforth referred to as "at infinity"), an object with motion retrograde to the BH's rotation must still rotate in the same direction as the ergosphere (Komissarov 2004). However, to conserve momentum, at infinity, the object must have some negative angular momentum to conserve, and thus, by analogy, it has negative energy at infinity (Comisso and Asenjo 2021). More generally, the "negative energy at infinity" represents a way to conserve angular momentum/energy within the system, by introducing the energy of the system as it is observed at an infinite distance away from the system (Koide and Arai 2008; Comisso and Asenjo 2021). When particles with this "negative energy" fall into the BH, like in the Penrose Process, the BH loses rotational energy and a particle with higher positive energy, than the original, escapes to infinity (due to energy conservation), causing energy extraction (Blandford and Znajek 1977; Piran and Shaham 1978).

Cross-applying this idea to the BZ mechanism, instead of a particle, we will be working with the electromagnetic (EM) field, as that is what sustains the negative energy at infinity (Koide and Arai 2008). In this case, the (force-free) magnetosphere is initially stationary, and so what was previously described above with the particles is also applicable to the field, except the matter is replaced by EM field lines, as long as $0 < \Omega_{\square} < \Omega_{\square}$, where $\Omega_{\square}$ and $\Omega_{\square}$ are the angular velocities of the BH and magnetic field respectively (Koide and Arai 2008). This condition makes intuitive sense as the way negative energy at infinity is produced, for the field lines, is because of the aforementioned process caused by frame-dragging: When the parts of the magnetic field lines closer to the EH are dragged more and the field twists, magnetic tension works to redistribute that angular momentum outwards through the rest of the field line (Blandford and Znajek 1977; Kinoshita and Igata 2018). What this means is that angular momentum is transported away from the BH, but also that the parts of the field getting more strongly frame-dragged are also losing some of the angular momentum they had as a result of the frame-dragging (Blandford and Znajek 1977; Kinoshita and Igata 2018). This becomes important, because that angular momentum was keeping that part of the field "still" in reference to the corotation vector (Kinoshita and Igata 2018). However, the transport of angular momentum by magnetic tension away from this portion of the field produces its negative energy at infinity due to having reduced angular momentum relative to the corotation vector (Koide and Arai 2008; Kinoshita and Igata 2018). As such, due to angular momentum redistribution doing the above, the magnetic field line obtains negative energy at infinity due to the frame of reference being the rotating ergosphere (Koide and Arai 2008; Comisso and Asenjo 2021; Liu 2022).

As these EM field lines are carried into the SMBH, the negative energy at infinity which they carry extracts energy from the SMBH (Kinoshita and Igata 2018). While energy extraction

from the SMBH seems contradictory to the definition of the EH, it makes sense since the energy extraction can be thought of as the negative energy at infinity of the magnetic field lines reducing the angular momentum of the SMBH (through the relation between angular momentum and energy within the ergosphere), and thus the angular momentum/energy of the SMBH thus decreases (Koide and Arai 2008; Kinoshita and Igata 2018). Additionally, under force-free conditions, where the Alfvén speed approaches the speed of light (as conditions approach the force-free limit) (Koide and Arai 2008), the Alfvén surface approaches the inner LS, which acts as a causal boundary for the Alfvén waves (Noda and Takahashi 2025). As such, when the magnetic field lines carry negative energy to the EH, the associated PF flows out from just outside the EH along the same field lines towards infinity (Koide and Arai 2008). This can be upheld because this energy is transported along the field lines in the form of tension, propagating outwards (Blandford and Znajek 1977; Kinoshita and Igata 2018). However, since the Alfvén surface at the inner LS is a purely ingoing causal boundary, the Alfvén waves and their information are causally restricted at the inner LS (Noda and Takahashi 2025). As such, the Alfvén surface being at the inner LS means that the PF outflow doesn't violate the EH's definition.

This perspective of looking at the BZ process also makes the concept of spinning up vs down more intuitive. Under the BZ framework, the energy extraction from the BH's rotational energy only occurs if the BH is rotating faster than the field lines since the field lines moving slower than the corotation vector is what extracts energy from the BH in the first place (Kinoshita and Igata 2018; Blandford and Znajek 1977; Koide and Arai 2008). This is what is known as the BH spinning down, and when $0 \leq \Omega_{\square} \leq \Omega_{\square}$ that means that the field lines extract rotational energy (Kinoshita and Igata 2018; Blandford and Znajek 1977). If the reverse is true, however, and the field lines are rotating faster than the BH is, $\Omega_{\square} > \Omega_{\square}$, then the BH begins to spin up since the field lines drag it, rather than vice-versa (Kinoshita and Igata 2018; Blandford and Znajek 1977). The result of this is that the BH extracts rotational energy from the field lines, which is the reverse of the BZ process (Kinoshita and Igata 2018; Blandford and Znajek 1977). With the understanding of negative energy at infinity, we can now gain a more nuanced understanding of this process. When the "switch-on" condition (simplified down)

$$\widehat{\square} \propto -(\square_{\square} - \square_{\square})$$

of the BZ process is active, where $\widehat{\square}$ represents the electromagnetic field, the energy of the field lines is negative at infinity (Koide and Arai 2008). As such, like discussed above the field lines reduce the BH's angular momentum/energy by crossing the EH (Koide and Arai 2008). If the reverse condition, $\Omega_{\square} > \Omega_{\square}$, were once again true the field lines would now have positive energy at infinity ($\square^{\infty}_{\square\square}$) due to the relation in Eq. [energyAtInfEq] (Koide and Arai 2008).

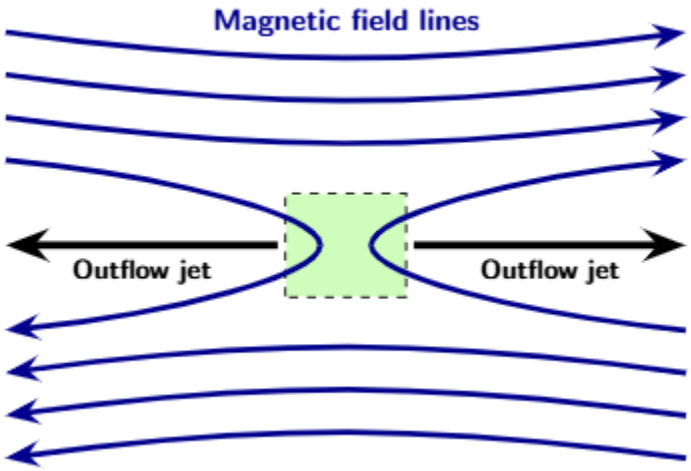
$$\square^{\infty}_{\square\square} \propto (\square_{\square} - \square_{\square})$$

As such, the field lines would cross the EH with positive energy, thus increasing the angular momentum of BH (Blandford and Znajek 1977; Kinoshita and Igata 2018).

3.4 Magnetic Reconnection

Given the understanding of how negative energy at infinity works, it is now possible to look at what MR entails. Before looking at MR in AGN, however, we must first understand the concept broadly.

When two magnetic field lines are running in opposite directions and both are being stretched extensively, it is possible for the field lines to disconnect from themselves and reconnect with the opposing field line in a diffusion region, while releasing massive amounts of energy in the process, as seen in Fig. 1 (Hesse and Cassak 2020; Yamada et al. 2010; Zweibel and Yamada 2016). A diffusion region is essentially where charged particles become decoupled from the magnetic field, and as such the magnetic field lines are able to reconnect since they are not tied to a physical object within this region (Beedle et al. 2025; Hesse and Cassak 2020). It is key to note that it is not possible for a magnetic field line to reconnect with itself, as no matter how much that field line is stretched, it will never disconnect since that would create a free end, which is not physically possible (Hesse and Cassak 2020). Hence, the field line can only change connectivity with another antiparallel magnetic field line (Hesse and Cassak 2020). In terms of why so much energy is released, in Hesse and Cassak (2020), a rubber band analogy is used to explain. Comparing the rubber band to a magnetic field line, as you stretch the band, you build up more and more potential energy within it. In other words, stretching the field rubber band stores energy within it. If you keep stretching it, however, at a certain point the rubber band ends up snapping, releasing all of the built up energy. Magnetic field lines behave much the same way, and so the analogy holds (Hesse and Cassak 2020). In the end, about half of the energy released by the magnetic field lines go into heating the plasma up (Hesse and Cassak 2020; Koide and Arai 2008), limiting the amount of energy actually accelerating the jet (Koide and Arai 2008; Komissarov et al. 2009).



Schematic of magnetic reconnection. Magnetic field lines run in opposite directions and reconnect to each other in the diffusion region (green). Adapted from Zweibel & Yamada (Zweibel and Yamada 2016).

After reconnection, as seen in Fig. 1, the resulting field lines are strongly bent near what was the reconnection site (Yi et al. 2018; Hesse and Cassak 2020). Again, it must be understood that the magnetic field lines being oriented as straight is a lower energy state for them (Myers et al. 2017). As such, tension works to straighten out these bent field lines, which then causes them to sling out like rubber bands (Hesse and Cassak 2020). Assuming the electric resistivity of the plasma is zero everywhere except in the diffusion region, along with the field lines, the plasma is coupled with the magnetic field, so it too gets flung out, causing two jets of plasma to form, moving in opposite directions away from the diffusion region (Koide and Arai 2008; Shen and YuChih 2025), something that will become key in the energy extraction mentioned later. But the last thing to note here is that MR is a self-driven process, which means that it causes itself to happen over and over again (Hesse and Cassak 2020). This is because the ejection of jets creates a low-pressure region where the surrounding plasma is brought into the region where plasma was previously accelerated out as a jet (Zweibel and Yamada 2016). With this accreted material, magnetic field lines follow, allowing for further reconnection to occur, enabling the process to happen over and over again as more and more accreted material is launched out (Hesse and Cassak 2020). This is another reason why MR is able to release such large amounts of energy: because it fuels itself to keep going (Hesse and Cassak 2020).

There are limits to MR's ability to create RJs, insofar as Joule Heating and finite length acceleration exist (Koide and Arai 2008). After the magnetic flux tubes (a bundle of magnetic field lines which form a tubular structure) reconnect and release massive amounts of energy, the terminal velocity of the plasma outflow is less than the theoretical maximum value (Koide and Arai 2008; Hesse and Cassak 2020). In terms of Joule Heating, some of the energy is dissipated away as heat in the diffusion region (Matteo 1998; Zweibel and Yamada 2016), instead of being used to further accelerate the jets. Then with finite length acceleration, there exists a limited spatial extent over which the plasma is accelerated by the MR, and so that limits the amount of released energy that can be converted into kinetic energy for the jets (Koide and Arai 2008).

Understanding this, we can now delve into why MR is an important modification to the BZ process. It has been established in the previous section that the BZ process relies on a mechanism of negative energy at infinity being created due to the ergosphere, and that negative energy reduces the rotational energy of the BH as positive energy is released (Koide and Arai 2008; Piran and Shaham 1978; Blandford and Znajek 1977). What MR proposes is the same idea of transporting negative energy, but differs in how such negative energy at infinity is established (Koide and Arai 2008). Whereas the classic BZ process relies on magnetic tension to lead to the redistribution of energy to produce such an effect (Blandford and Znajek 1977), MR can also create the same condition of negative energy at infinity in a different way (Koide and Arai 2008).

From earlier, we reached the conclusion that MR creates two outflows which move in opposite directions. The speed of those jets are represented by $v_{out} = v'_{out}$ and $v_{out} = -v'_{out}$, where v_{out} is the plasma outflow velocity and v'_{out} is the estimated outflow velocity assuming all the magnetic energy (from MR) is converted into kinetic energy (Koide and Arai 2008). The signs in the above equations also indicate whether the

jet is moving prograde or retrograde to the BH's rotation (with $-\square'_{\square\square\square}$ being retrograde) (Liu 2022). The energies of those jets at infinity are represented by $\epsilon^{\infty}_{\pm}$, and the $\pm$ subscript once again refers to direction of velocity with respect to the BH's rotation (Koide and Arai 2008). It is then seen that if $\epsilon^{\infty}_{-} < 0$, then that means that the plasma outflow moving retrograde to the BH's rotation is moving with negative energy at infinity (Koide and Arai 2008). The condition for determining the sign of ϵ^{∞}_{-} (due to MR) is dependent on the relation $\square_{\square} > \square^{-}_{\square}$ (Koide and Arai 2008). From this, $\square_{\square}$ represents the Alfvén (four) velocity and $\square^{-}_{\square}$ is the critical Alfvén four velocity for the BH (Koide and Arai 2008). It is key to note that $\square_{\square}$ itself is directly dependent on magnetic field strength ($\square$). Hence, in accordance with the relation from Eq. [alfvenVelocity], a higher $\square_{\square}$ indicates a stronger magnetic field, and a stronger magnetic field releases more energy during MR since magnetic field energy density ($\square_{\square}$) follows the relation in Eq. [magDensity].

$$\square_{\square} \propto \square^{2}$$

Recontextualizing all of the above, ϵ^{∞}_{-} only becomes negative after the magnetic field becomes strong enough (when $\square_{\square} > \square^{-}_{\square}$) that the MR creates a retrograde outflow that has enough retrograde angular momentum to have negative energy at infinity (Koide and Arai 2008; Liu 2022). This plasma outflow will eventually fall into the BH where it will decrease the BH's rotational energy and overall energy (Koide and Arai 2008; Liu 2022; Comisso and Asenjo 2021).

While this is happening, the other jet is moving in a direction of the BH's rotation, or in a prograde direction (Koide and Arai 2008; Comisso and Asenjo 2021). For this outflow, $\Delta\epsilon^{\infty}_{+}$ is positive if $\square_{\square}$ is large enough, with the condition closely mirroring that of the retrograde jet's: $\square_{\square} > \square^{+}_{\square}$ (Koide and Arai 2008). Again, all that this condition is saying is that as the magnetic field gets stronger, more energy is released by MR, and so the prograde jet ends up accelerated to where the jet can escape to infinity (Koide and Arai 2008; Liu 2022; Comisso and Asenjo 2021). As such, with the retrograde jet we end up reducing the amount of rotational energy that the BH has, while with the prograde jet, on net more energy escapes along magnetic field lines stretching outwards to infinity, thus causing energy extraction via MR (Koide and Arai 2008; Liu 2022; Comisso and Asenjo 2021).

In this way, the process of MR manifests as a way to redistribute angular momentum within the plasma in order to create the negative energy at infinity which extracts BH rotational energy (Koide and Arai 2008; Liu 2022; Comisso and Asenjo 2021). Since angular momentum (which is coupled with energy) counter-rotational to the BH (hence negative energy at infinity) is given to the retrograde outflow, that co-rotational angular momentum must be given to the prograde outflow, thus allowing for energy to escape the system with the prograde jet, while the retrograde jet falls into the BH, reducing its energy/angular momentum (Koide and Arai 2008; Liu 2022; Comisso and Asenjo 2021).

As the two jets continue to be separated, the magnetic flux tubes which connect both of them get stretched in opposite directions and bent by the Ergosphere (Koide and Arai 2008; Shen and YuChih 2025). This leads to repeated conditions for reconnection to occur, and so repeated reconnection ends up occurring and more energy is extracted from the BH (Koide and Arai 2008; Comisso and Asenjo 2021).

3.5 Comparative Analysis

While the BZ and MR mechanisms are similar in that they both use the magnetic field to extract energy from the SMBH powering the AGN, they differ in how they actually operate. As discussed in this review, both mechanisms fundamentally rely on negative energy at infinity flowing into the SMBH to reduce the BH's "reducible" mass, thus allowing energy extraction. However, the BZ process has been shown to be continuous with its extraction of rotational energy from the BH, while the MR process relies on more unstable events and sudden "bursts" of energy extraction to occur. MR is also shown to be more efficient with energy extraction compared to the BZ mechanism (Shen and YuChih 2025; Comisso and Asenjo 2021). Even considering these differences, despite the fact that many sources characterize them as alternative explanations of the same concept, the reality behind these energy extraction mechanisms may be found by considering MR and the BZ mechanism to be complementary to each other (Singh et al. 2018). This is further reinforced by how the BZ mechanism, as shown above, can work under the same framework that MR operates under. In other words, while both mechanisms extract energy in different ways, it may be more likely that both operate simultaneously to extract energy from the AGN, rather than as substitutes for one another. According to Tchekhovskoy et al. (2011) and Singh et al. (2018), the BZ mechanism and MR have jet efficiencies of $\eta = 30\%$ and $\eta = 150\%$, respectively, thus characterizing the BZ mechanism as a continuous, but slower energy extraction mechanism. On the other hand, the information in this review can be synthesized to thus characterize MR as a much more intense, yet intermittent form of energy extraction. In this way, we can begin to think of them as building upon each other rather than competing as mechanisms when it comes to launching RJs as both extract energy from the SMBH in different, yet potentially complementary ways.

4 Conclusion

Throughout this review, the BZ mechanism has been described and then recontextualized with the same negative energy at infinity characterization which MR carries. Just like the BZ mechanism represented a new way of looking at BH energy extraction as compared to the Penrose Process, MR now offers a new explanation. This time, however, the mechanisms are not necessarily mutually exclusive, rather complimenting each other more than they seem to negate one another. As such, it may very well be possible that RJs are not just powered by one of these mechanisms, but a combination of both. Figuring out the intricacies behind the interaction between these two mechanisms is still a field which is actively developing, and understanding that dynamic is what our future work will be centered around, whilst in pursuit of a greater understanding of the astrophysical phenomena which shape our universe.

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